



M000 Week 0 — English Worksheet

Topic: Pixel Art — Counting Squares · **Course:** Builder Basics · **Type:** Warm-Up (before Week 1) ·

Time: about 40 minutes

Every block in Minecraft has three numbers: **X** (left ↔ right), **Y** (up ↓ down), and **Z** (depth, in ↔ out). Pixel art sits on a flat wall, so **Z stays the same** the whole time — you only change **X** and **Y**.



This is a quick warm-up to do **before Week 1**. It teaches you to count squares so your builds line up perfectly.

1 · Read the Wall 🔍

Open the **Pixel Art** world. In front of you is a flat wall with red pixel art on it.

Think of the wall like graph paper:

- **X** counts the columns, left to right →
- **Y** counts the rows, bottom to top ↑
- **Z** is how deep the wall is. It is the **same** for every red square, so we can ignore it here.

Every red shape you build needs two corners: a **START** and an **END**. The rest of this worksheet is one skill — **how to count to those two corners**.

Look at the wall and find the smallest red mark — a single square. Do you think its START and its END are the same spot, or two different spots?

2 · Find START and END — Step by Step 🗺️

Here is the recipe. Use it for **every** straight line or square.

① **Count X across** (left → right): 0, 1, 2, 3 ... Find the column where the red **begins** and where it **ends**. ② **Count Y up** (bottom → top): 0, 1, 2, 3 ... Find the row where the red **begins** and where it **ends**. ③ **START = (left X, bottom Y). END = (right X, top Y)**. ④ **Same-number check**: 1 square **wide** → START X = END X. 1 square **tall** → START Y = END Y.

In the grids below: ■ **START** · ■ **END** · ■ **in between** · ■ **START and END are the same square**.

Case 1 — A single square

Y \ X	0	1	2	3	4
3	■	■	■	■	■
2	■	■	■	■	■
1	■	■	■	■	■
0	■	■	■	■	■

- ① **Count X**: 0, 1, 2, **3** → red begins at X 3 and ends at X 3.
- ② **Count Y**: 0, 1, **2** → red begins at Y 2 and ends at Y 2.
- ③ **START = (3, 2). END = (3, 2)**.
- ④ 1 wide **and** 1 tall → START and END are the **same square**. ■

Blocks

```
fill [ ■ red wool ]  
  from world X 3 Y 2 Z 0  
  to world X 3 Y 2 Z 0
```

Python

```
blocks.fill(RED_WOOL, world(3, 2, 0), world(3, 2, 0), FillOperation.Replace)
```

💡 For a single block you can also use `blocks.place(RED_WOOL, world(3, 2, 0))` — place needs only **one** spot, because there is nothing to count.

Case 2 — A line lying down (1 tall)

Y \ X	0	1	2	3	4
3					
2					
1					
0					

- ① **Count X:** red fills columns **1, 2, 3** → begins at X 1, ends at X 3.
- ② **Count Y:** red is only in row **2** → begins at Y 2, ends at Y 2.
- ③ **START = (1, 2). END = (3, 2).**
- ④ 1 tall → **START Y = END Y = 2** (same Y!).

Python

```
blocks.fill(RED_WOOL, world(1, 2, 0), world(3, 2, 0), FillOperation.Replace)
```

Case 3 — A line standing up (1 wide)

Y \ X	0	1	2
3			
2			
1			
0			

- ① **Count X:** red is only in column **1** → begins at X 1, ends at X 1.
- ② **Count Y:** red fills rows **1, 2, 3** → begins at Y 1, ends at Y 3.
- ③ **START = (1, 1). END = (1, 3).**
- ④ 1 wide → **START X = END X = 1** (same X!).

Python

```
blocks.fill(RED_WOOL, world(1, 1, 0), world(1, 3, 0), FillOperation.Replace)
```

🔑 **The big idea:** when a shape is **1 square thick**, one number does **not** change — START and END share it. That is correct, not a mistake.

3 · Your Turn: Find START and END

Use the recipe (count X across, count Y up). For each shape, write START, END, and circle the number that is the **same**.

Shape A

Y \ X	0	1	2	3
2				
1				
0				

START = (__ , __) END = (__ , __) Same number: X · Y · both

Shape B

Y \ X	0	1	2	3
3				
2				

START = (__ , __) END = (__ , __) Same number: X · Y · both

Shape C

Y \ X	0	1	2	3
2				
1				
0				

START = (__ , __) END = (__ , __) Same number: X · Y · both

4 · Count the Length 1234

Counting squares has a trap. From **X = 2** to **X = 5** is **not** 5 squares — it is **2, 3, 4, 5 = 4 squares**. You count the START square too.

Length = (END – START) + 1. So to make a line **N** long starting at X, the END is **X + (N – 1)** — not X + N.

You want a line lying down that is 3 squares long, starting at X = 1 (Y stays 2). What is the END X?

Now write the fill for it — blocks or python, your choice:

You want a line standing up, 5 squares tall, starting at Y = 0 (X stays 6). What is the END Y?

5 · Spot the Bug 🐛

Each command was meant to do one thing, but it is wrong. Rewrite it so it works, then explain why the original was wrong.

Bug A — Meant to color **one** square at X 4, Y 4.

```
blocks.fill(RED_WOOL, world(4, 4, 0), world(5, 5, 0), FillOperation.Replace)
```

Hint: count how many squares (4,4) to (5,5) really covers.

Write the fixed command:

Why was it wrong? Why does your fix work?

Bug B — Meant to make a line lying down **3** squares long starting at X 0 (Y stays 1).

```
blocks.fill(RED_WOOL, world(0, 1, 0), world(3, 1, 0), FillOperation.Replace)
```

Write the fixed command:

Why was it wrong? Why does your fix work?

Bug C — Meant to make a line **standing up**, 4 tall at X 2, starting Y 0.

```
blocks.fill(RED_WOOL, world(2, 0, 0), world(5, 0, 0), FillOperation.Replace)
```

Hint: for a standing line, which number should change — X or Y?

Write the fixed command:

Why was it wrong? Why does your fix work?

6 · Build It 📷

Switch to your homework world. **Build this red picture** on the wall — it is the same art you can see in the world. **X** runs across, **Y** runs up.

Y \ X	0	1	2	3	4	5	6	7	8	9	10
10	■	■	■	■	■	■	■	■	■	■	■
9	■	■	■	■	■	■	■	■	■	■	■
8	■	■	■	■	■	■	■	■	■	■	■
7	■	■	■	■	■	■	■	■	■	■	■
6	■	■	■	■	■	■	■	■	■	■	■
5	■	■	■	■	■	■	■	■	■	■	■
4	■	■	■	■	■	■	■	■	■	■	■
3	■	■	■	■	■	■	■	■	■	■	■
2	■	■	■	■	■	■	■	■	■	■	■
1	■	■	■	■	■	■	■	■	■	■	■
0	■	■	■	■	■	■	■	■	■	■	■

Work **one shape at a time**. For each red shape: **find its START and END** (count X across, then Y up), check if a number is the **same** (1 wide or 1 tall), then **fill** it. Easy ones first — the single squares, then the lines, then the big block.

Send a photo or video of your wall, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

My first square had START at X ... , Y ... and END at the same spot because ...

To find START and END I counted ...

To make a line ... long, I counted ...

The hardest part was ...

To fix it, I ...

Next time I would ...

My first square had START at X ... , Y ... and END at the same spot because ...

To find START and END I counted ...

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7 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) while you build one square and one line. Talk through it like you are teaching someone who has never seen it. Try to use these words: **X, Y, START, END, same, fill**.

1. Show the empty wall and point out X (across) and Y (up).
2. Color one square. Count to its START and END out loud and say why they are the same.
3. Make a line. Count to its START and END out loud.
4. Show one counting mistake and how you fixed it.

Write what you will say in your video. Plan it below before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.